Numerical Analysis Assignment 5

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1 Problem 4.6

For the convergence test, I am using the following equation:

```
if np.all(np.abs(A_k - np.diag(np.diag(A_k))) < tolerance):
    return np.diag(A_k), iter
```

Figure 1: Convergence test

In this equation, if the values of the off diagonal entries come closer to 0, then the fucntion will stop iteration and will return the values.

```
Testing first:
Matrix:
[[1.e+00 1.e+03]
 [1.e-03 1.e+00]]

Our eigenvals:
Eigenvalues: [3.49942297e-13 2.00000000e+00]
Iterations needed: 1000

NumPy eigvals:
[ 2.00000000e+00 -2.08166817e-17]

Abs difference:
[3.49921481e-13 5.86197757e-14]
```

Figure 2: First Testing

```
Testing second:
Matrix:
[[   1. 1000.]
 [   0.    1.]]

Our eigenvals:
Eigenvalues: [1. 1.]
Iterations needed: 1000

NumPy eigvals:
[1. 1.]

Abs difference:
[0. 0.]
```

Figure 3: Second Testing

```
Testing third:
Matrix:
[[ 2.  3.  2.]
 [10.  3.  4.]
 [ 3.  6.  1.]]

Our eigenvals:
Eigenvalues: [11. -3. -2.]
Iterations needed: 1000

NumPy eigvals:
[11. -2. -3.]

Abs difference:
[6.66133815e-16 1.33226763e-15 5.32907052e-15]
```

Figure 4: Third Testing

We can see above that absolute difference for each of the tests are almost zero. The Matrices used for the tests are the following:

```python
A1 = np.array([[1, 1000],
               [0.001, 1]], dtype=np.float64)

B1 = np.array([[1, 1000],
               [0, 1]], dtype=np.float64)

A2 = np.array([[2, 3, 2],
               [10, 3, 4],
               [3, 6, 1]], dtype=np.float64)
```

Figure 5: Test Matrices

To find the QR decomposition we used Householder transformations and after iteration and finding the eigenvalues, we compared the results with the results of the eigenvalue results given by numpy.

2 Problem 4.7

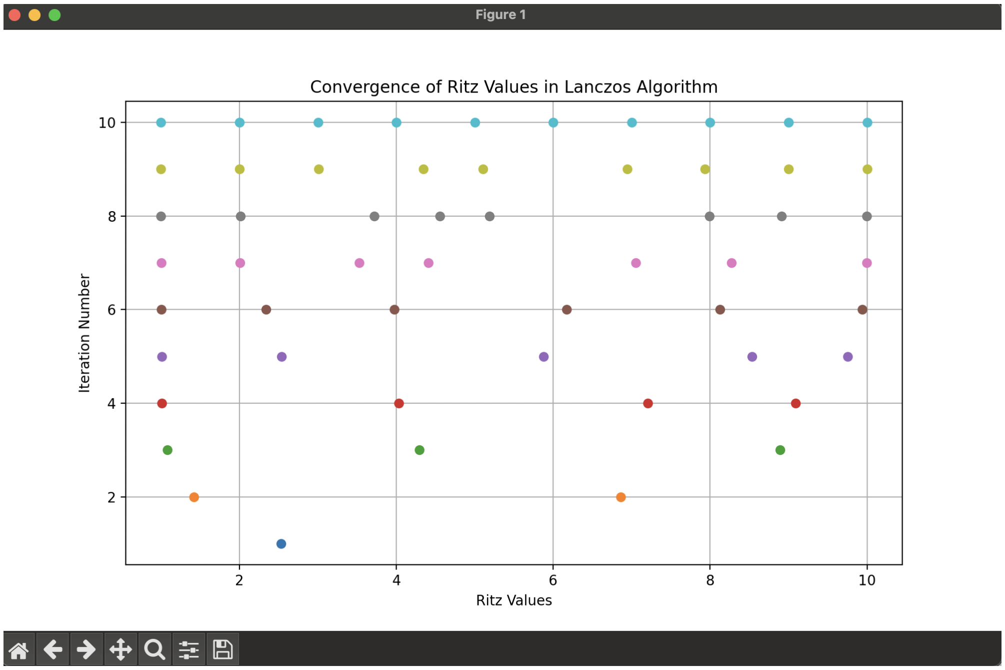

Figure 6: Iteration number vs Ritz values for n = 10

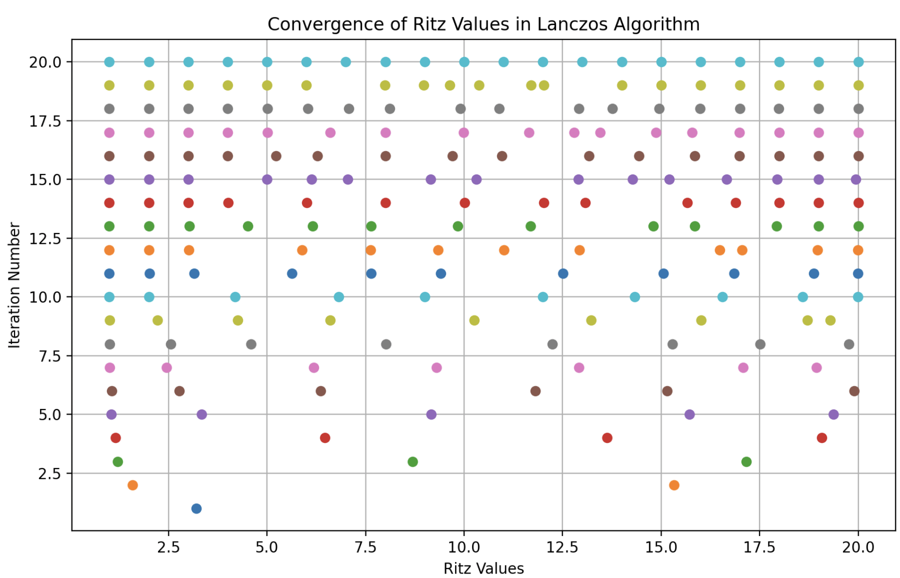

Figure 7: Iteration number vs Ritz values for n = 20

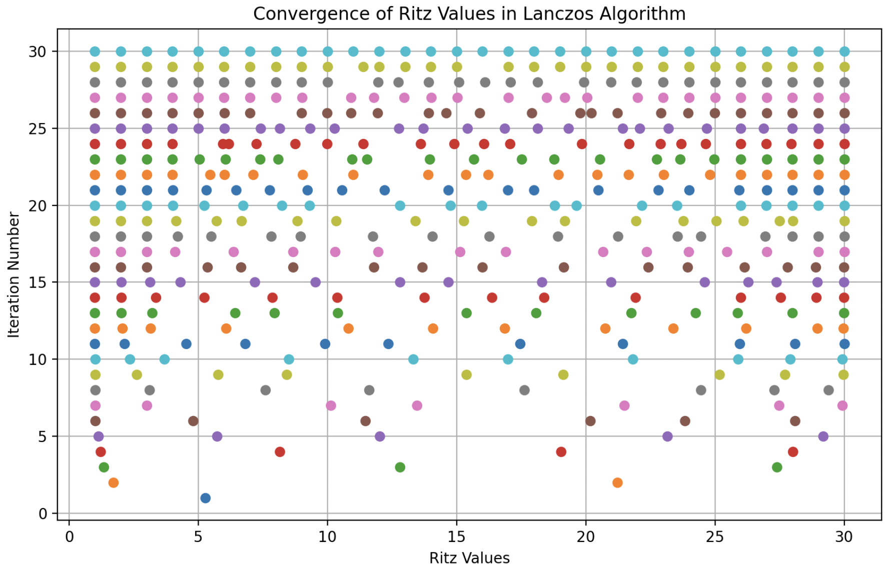

Figure 8: Iteration number vs Ritz values for n = 30

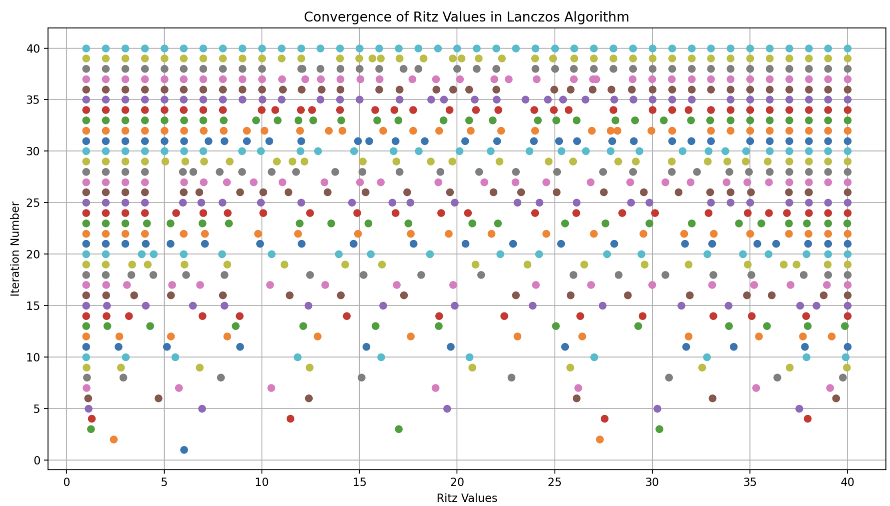

Figure 9: Iteration number vs Ritz values for n = 40

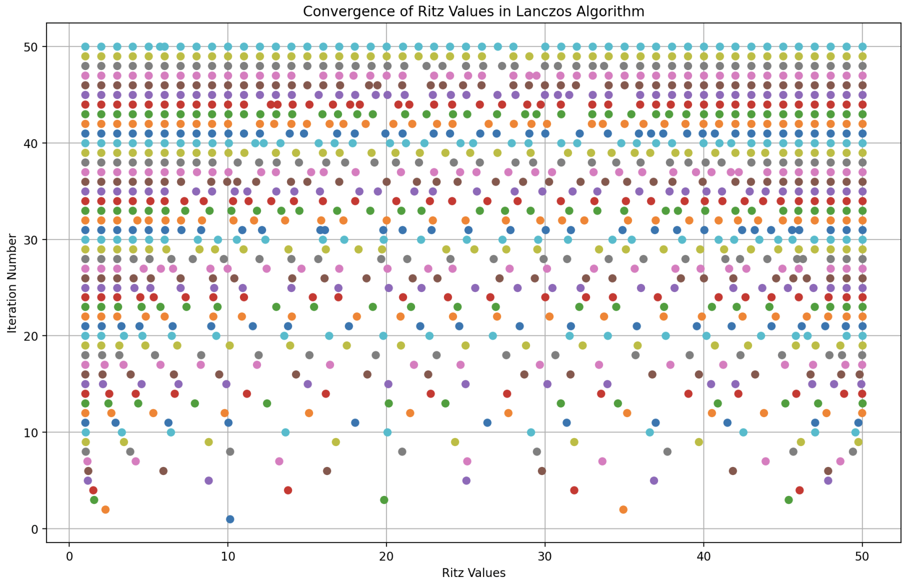

Figure 10: Iteration number vs Ritz values for n = 50

We can see that the Ritz values slowly converge to the final values as the iteration number increases. Also, we can see that for the 2nd iteration we only can find 2 Ritz values. But as we iterate more, the found number of Ritz values become more. Some new Ritz values emerge and they start to converge to the final values.

3 Problem 5.1

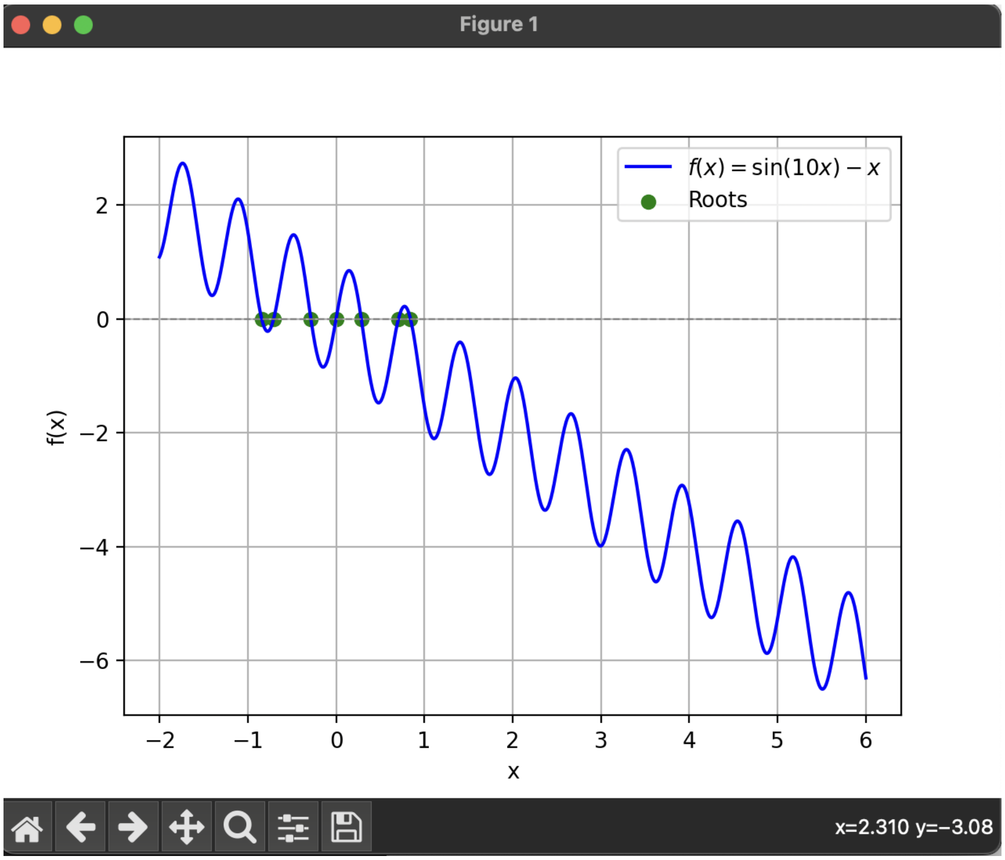

Figure 11: Roots on the graph

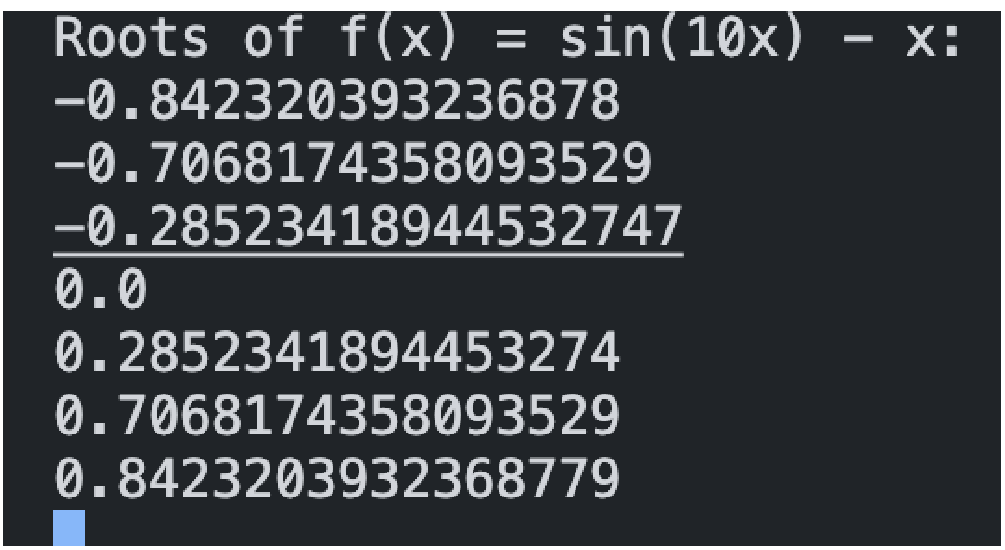
```
Roots of f(x) = sin(10x) - x:
-0.842320393236878
-0.7068174358093529
-0.28523418944532747
0.0
0.2852341894453274
0.7068174358093529
0.8423203932368779
```

Figure 12: Roots of the function

In this problem, we find the roots of the function f(x) = sin(10x) - x in the interval [-5, 5]. Moreover, plt.plot creates a plot of f(x) = sin(10x) - x in blue. plt.axhline (0, color='gray', lw=0.8, linestyle='–') draws a horizontal dashed line at y = 0 to represent the x-axis. plt.scatter plots each root found as a green dot at y = 0 (indicating roots where f(x) = 0).